📖 1. Room Overview

The **DNS in Detail** room on TryHackMe explores the **Domain Name System (DNS)**. DNS is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet. It translates human-readable domain names (like `google.com`) into machine-readable IP addresses (like `142.251.46.238`).

🏗️ 2. Task 1: What is DNS?

DNS exists so we don't have to memorize thousands of IP addresses. It functions exactly like a contact list on a phone.

📝 Key Definitions:

- **Domain Name:** The human-friendly address (e.g., `tryhackme.com`).
- **IP Address:** The actual location of the server on the internet.
- **TLD (Top-Level Domain):** The suffix of the domain, such as `.com`, `.org`, or `.net`.

ANSWERS:

What does DNS stand for?

Domain Name System

🌳 3. Task 2: The DNS Hierarchy

DNS is not a single server; it is a tree-like structure. When you look up a site, your request goes through several levels:

1. **Root Servers:** The top of the tree. They direct the query to the correct TLD server.
2. **TLD Servers:** Managed by organizations like IANA. They handle specific extensions (e.g., the `.com` server).

3. **Authoritative Servers:** The final stop. These servers hold the actual DNS records for a specific domain.

ANSWERS:

What is the maximum length of a subdomain?

63

Which of the following characters cannot be used in a subdomain (3 b _ —)?

_ (the third option)

What is the maximum length of a domain name?

253

What type of TLD is .co.uk?

ccTLD

🔄 4. Task 3: Record Types

To showcase your technical skills, it is important to understand the different types of DNS records used to direct traffic:

- **A Record:** Maps a hostname to an **IPv4** address.
- **AAAA Record:** Maps a hostname to an **IPv6** address.
- **CNAME (Canonical Name):** An alias. It points one domain to another domain (e.g., `blog.mysite.com` -> `mysite.com`).

- **MX Record (Mail Exchange):** Directs email to the correct mail server.
- **TXT Record:** Used for verification (like SPF for email security) and general information.

ANSWERS:

What type of record would be used to advise where to send email?

MX

What type of record handles IPv6 addresses?

AAAA

🛠️ 5. Task 4: Making a Request (The Recursive Process)

When you type a URL into your browser, the following happens:

1. **Recursive Resolver:** Usually your ISP's server. It does the "legwork" to find the IP.
2. **Iterative Queries:** The resolver talks to the **Root**, then the **TLD**, then the **Authoritative** server.
3. **Caching:** Your computer saves the result locally so it doesn't have to ask again for a set period (the **TTL** or Time To Live).

ANSWERS:

What field specifies how long a DNS record should be cached for?

TTL

What type of DNS Server is usually provided by your ISP?

Recursive

What type of server holds all the records for a domain?

Authoritative

🚩 6. Lab Activity & Flags

In this task, I used tools like `nslookup` to query servers manually.

Task 5

What is the CNAME of shop.website.thm?

shops.myshopify.com

What is the value of the TXT record of website.thm?

THM[redacted]

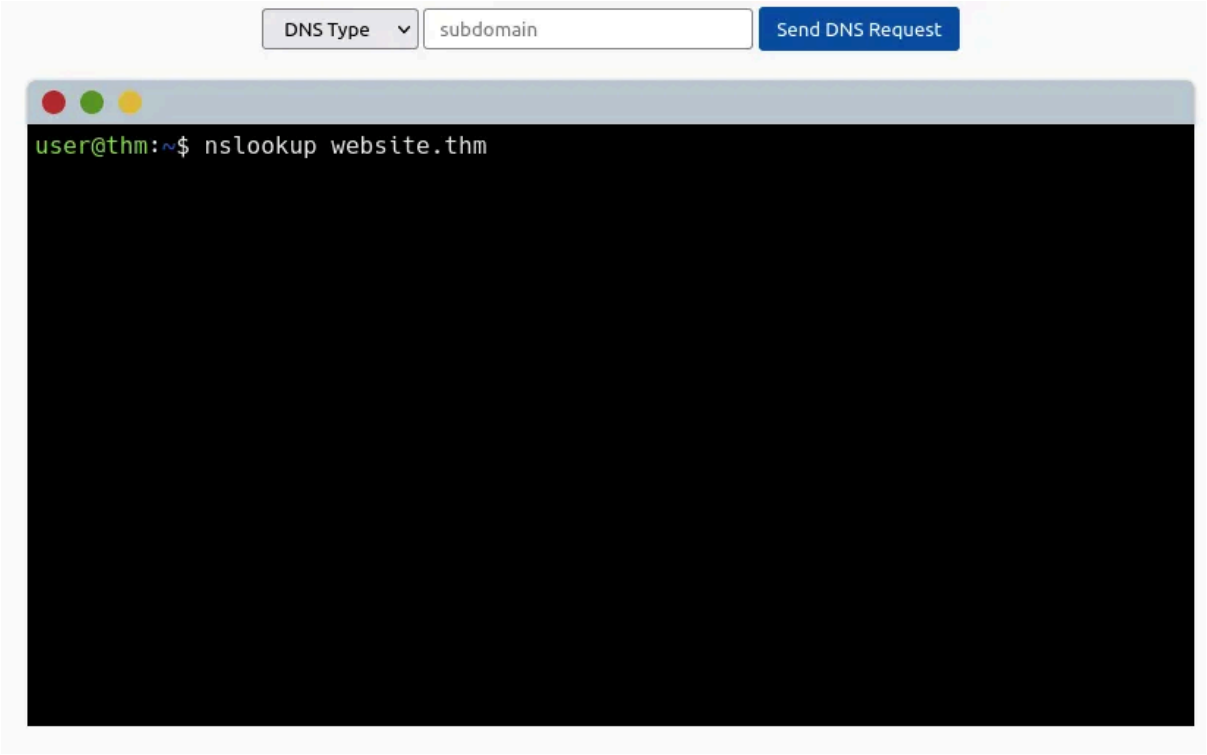
What is the numerical priority value for the MX record?

30

What is the IP address for the A record of www.website.thm?

10.10.10.10

📸 Lab Evidence




```
user@thm:~$ nslookup --type=CNAME shop.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
shop.website.thm canonical name = shops.myshopify.com
```

```
user@thm:~$ nslookup --type=TXT website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm text = "THM{7012BBA60997F35A9516C2E16D2944FF}"
```

```
user@thm:~$ nslookup --type=MX website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm mail exchanger = 30 alt4.aspmx.l.google.com
```

```
user@thm:~$ nslookup --type=A website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: website.thm
Address: 10.10.10.10
```

🎓 7. Conclusion & Skills Gained

By completing this lab, I mastered:

- Using **nslookup** and **dig** to troubleshoot DNS issues.
- Identifying different **DNS Record types** and their purposes.
- Understanding the **Recursive vs. Iterative** lookup process.